

The development and validation of an early warning system to prevent the acquisition of 12-lead resting ECGs with interchanged electrode positions

Johan de Bie, PhD,* David W. Mortara, PhD, FACC, Todd F. Clark

Mortara Instrument Inc., 7865 North 86th Street, 53224 Milwaukee, WI, USA

Abstract

Background: A fraction of routine resting ECG's are taken with electrode positions interchanged, leading to possible clinical misinterpretation.

Objective: Develop and test a method to detect and prevent electrode reversals at the electrocardiograph before the ECG is acquired.

Method: The algorithm is based on QRS axis and P amplitudes for limb electrode reversals, and P-Q-RS amplitude distances to detect chest electrode reversals. The evaluation method involved a large (>18,000) hospital database for which serial ECG's were available and was based on simulated juxtapositions.

Results: The 7 most common lead reversals could be detected with a specificity of 99.8% per type and an average sensitivity of 90%, excluding LA-LL reversal (22% sensitivity).

Discussion: Results are similar to retrospective studies that used smaller, more homogeneous datasets.

Conclusion: The early warning system reduces the ECG's recorded with reversal by 80%, at the price of a modest false alert rate of 1.4%.

© 2014 Elsevier Inc. All rights reserved.

Keywords:

Electrocardiogram; ECG; Electrode reversal; Electrode interchange; Electrode misplacement; Automatic detection

Introduction

Correct electrode positions are important for the interpretation of 12-lead resting ECGs. The most frequent position errors are those of the chest electrode with respect to the anatomical landmarks of the body and juxtapositions of electrode leads. This work is about the latter; it describes a method to detect lead reversals in real time and to evaluate its performance.

The frequency of occurrence of electrode juxtapositions is not well known; estimates vary between 0.4% and 4% of all ECGs taken [1]. Interchanges occur between limb electrodes (e.g. left arm versus right arm) and in the order of the chest leads (e.g. V2, V1, V3 instead of V1, V2, V3). Mistaking chest lead wires for limb lead wires is less likely because of color coding and the usually different length of the wires.

Many ECGs with lead reversals are not detected by routine interpretations, giving rise to possible inaccurate diagnoses [2,3]. Computer interpretation programs traditionally only attempt to detect a right arm–left arm switch (although more electrode reversal cases have been added

recently), but do not attempt to “correct” this situation, and only give a warning.

Several algorithms for retrospective recognition of electrode interchanges have been published. Hedén and coworkers [4] developed and tested a method based on artificial neural networks, first only for right arm–left arm reversals, subsequently also for other limb and chest electrode reversals [5]. Kors and Van Herpen [6] developed a method based on the correlation between the taken ECG lead and its reconstruction from the other 7 independent leads. Xia et al. [7] retested both of these algorithms on different data and also developed a method that combined criteria from both.

The objective of our study is to develop and test a method to reduce the number of ECGs taken with lead reversals by giving an alert to the technician for possible lead reversals before the ECG is finalized.

Materials and methods

Algorithm

During the observation period, when electrodes are connected and the ECG technician is waiting for the signal to stabilize, a running average of the P–QRS–T complex is maintained over 10 seconds. This complex is constantly analyzed for a lead reversal, and a warning is given on the

* Corresponding author at: Mortara Instrument Europe s.r.l., Via Cimarosa 103, 40033 Casalecchio di Reno, Italy.

E-mail address: johan.debie@mortara.com

screen of the electrocardiograph to check the electrode positions if a possible lead reversal is detected, in a manner similar to the warning that an electrode is not connected. The ECG technician has the possibility to ignore the warning and take the ECG anyway. When the technician presses the “Take ECG” button, the last 10 seconds is analyzed and stored. In this manner, the lead reversal detection algorithm does not add any time to the standard procedure of taking ECGs.

The algorithm has been developed to detect the reversals described in Table 1.

No attempt has been made to detect right leg–left leg reversals, because the effect on the ECG of this reversal is minimal and clinically insignificant. Also, non-neighbor chest lead reversals are not considered; they would trigger a neighbor reversal detection anyway, and the purpose of the algorithm is to detect a switch, not to precisely identify it.

Because of the low prior probability of a switch, the algorithm has been designed to have a high specificity while maintaining a reasonable sensitivity. The objective is to detect the majority of the lead reversals, without giving too many nuisance messages that would interrupt the ECG technician’s work flow unnecessarily, and potentially lead to the reversal messages being ignored.

ECGs with a ventricular pacemaker rhythm are ignored by the algorithm. The absence of the P-wave and the unusual QRS morphology of many pacemaker ECGs would lead to more false positives than justified for the purpose, considering the low percentage of pacemaker ECGs in routine clinical practice.

The detection of RA–LA lead reversal relies on QRS axis, P amplitude, and lead I and V6 being very dissimilar. Similar, albeit less sensitive rules were developed for detection of LA–LL and RA–LL reversals. For precordial leads, a matrix was calculated of the P, Q, R, and S amplitudes and additional waveform data for each lead. The distances between the leads in this multidimensional space were then calculated. Call this distance D_{ij} . In normal conditions, that is, no electrode interchange, you would expect the distance value between leads from neighboring electrodes to be the smallest. So the rule for, say V3–V4 exchange detection is $D_{23} > D_{24}$ and $D_{45} > D_{35}$. In other words, something is wrong if V2 appears to be closer to V4 than it is to V3 and simultaneously, V5 appears to be closer to V3 than to V4. However, if V3 and V4 are interchanged these inequalities are to be expected. Obviously, only half of

this rule can be used for the V1–V2 and V5–V6 reversals since there are only neighbors at one side.

Validation

The algorithm was evaluated using a large database of recorded ECGs from a medium size university hospital. The database contained circa 250,000 ECGs of a mix of outpatients and hospitalized patients that were recorded between February 2003 and April 2012. The database was cleansed on the basis of non-ECG parameters to remove test records, unidentified patients and patients with discrepancies or idiosyncrasies in demographic data (e.g. non-matching ID number and name in serial ECGs, unusual IDs, names with non-alphanumeric characters).

Subsequently, patients were selected who had at least three ECGs in the database that were taken on different days, forming three groups of ECGs, circa 20,000 ECGs each. These ECGs were exported from the database. All ECGs were then analyzed for the presence of decoding errors, missing leads and the presence of a “pacemaker” call in the automatic interpretation. If any of these conditions occurred, all three ECGs of that patient were removed. A total number of 18,654 ECG triplets remained for final processing. The ECGs were processed by the same lead reversal algorithm as in real time on the electrocardiograph. The ECGs from the database were 10 seconds in length, the same duration used in real time for the construction of the running average P–QRS–T complex. Of the three groups, one was used for performance analysis, and the other two were used for comparison as explained below.

In order to assess the specificity of the algorithm, it is necessary to separate the true from the false positives. True positive lead reversals were found in a two-stage process. First, all cases with positive lead reversals for which a reversal was also found in one or both of the comparison ECGs of the same patient were considered false; it is highly unlikely that a real electrode reversal would have occurred twice on different days. Then, all other positive cases were reviewed by an expert. The two serial ECGs were presented together with the test ECG as an aid to confirm the expert’s judgment, which was particularly helpful for limb lead reversals. Using the serial ECGs, most of the time there was little doubt whether a reversal had taken place, as any experienced ECG reader will confirm. Whenever the reviewer was in doubt, the call was considered to be false, based on the low prior probability of a true reversal. A limb lead reversal call was considered true also if the reviewer was of the opinion that a different limb reversal had taken place than called by the algorithm, e.g. if the algorithm called RA–LA reversal, but the reviewer called it RA–LL reversal, the case was considered true. The same method was followed for chest lead reversals. A total of 230 ECGs with an electrode reversal were found in this way. Note that these were not the only ECGs in the database with a true reversal, only those that the algorithm found and were confirmed.

Subsequently, all true positive lead reversals were removed, resulting in 18,424 ECGs, which now did not include any true reversals that were detectable by the algorithm. In order to assess the sensitivity of the algorithm,

Table 1
Table of electrode reversals taken into account by the algorithm.

Limb leads	Chest leads
Right arm–left arm: RA–LA (R–L)	V1–V2 (C1–C2)
Left arm–left leg: LA–LL (L–F)	V2–V3 (C2–C3)
Right arm–left leg: RA–LL (R–F)	V3–V4 (C3–C4)
Left arm–right leg: LA–RL (L–N)	V4–V5 (C4–C5)
Right arm–right leg: RA–RL (R–N)	V5–V6 (C5–C6)
Right arm–right leg + left arm–left leg: Ar–Lg	

In parentheses are the IEC denominations of the electrodes. Throughout the article, the AHA denominations will be used.

electrode reversals were now simulated by modifying or switching the ECG leads appropriately: e.g. to simulate an RA–LA reversal, lead I was inverted and leads II and III were switched. See Ref. [1] for a description of how each electrode reversal affects the ECG leads, and how they can be simulated. The algorithm was run again with one introduced reversal at a time, and the number of reversal calls was counted to obtain the program's sensitivity.

Results

The summary of the results can be found in Table 2 for limb electrode reversals and Table 3 for chest electrode reversals.

False positives are the positive cases from the original dataset that were considered false either by the reviewer or because one or both of the serial ECGs were also positive. True negatives are all cases in which the program did not find a reversal. True positive is the count of reversals found when a simulated reversal was introduced. Finally, false negative is the number of cases missed when a reversal was simulated. “True switch” indicates the number of lead reversal cases that were found in the original set, were found true and were removed from the set.

Sensitivity is calculated as $TP/(TP + FN) * 100\%$, specificity is calculated as $TN/(TN + FP) * 100\%$.

The algorithm was run on the original (not switched) dataset also for the lead reversals that involved the right leg (neutral) electrode, but clearly a switch could not be simulated since the right leg electrode is not involved in constructing the ECG leads. No false positives were found and the few true positive cases were removed from the set. Sensitivity and specificity could not be calculated for the switches that involve the right leg, obviously. A switch between right leg and left leg electrodes has little or no effect on the ECG and was ignored.

Discussion

The algorithm functions very well for chest lead reversals. Overall, 89% of the reversals are detected with a specificity of 99.80%. Even the most difficult switch, between V1 and V2 had a 75% sensitivity. It is interesting to note that true V1–V2 and V3–V4 reversals are relatively rare in this dataset, while V5–V6 reversals are much more common than the average. This is exactly the same finding as in Ref. [5].

Right arm–left arm reversal detection has a performance similar to the chest lead reversals. Left arm–left leg reversal

Table 2

Results of the algorithm for limb electrode reversals; see text for explanation of the legends.

	RA–LA	LA–LL	RA–LL	LA–RL	RA–RL	Ar–Lg
True negative	18,375	18,384	18,319	18,424	18,424	18,424
False positive	49	40	105	0	0	0
True positive	16,750	4134	9320			
False negative	1674	14,290	9104			
Sensitivity (%)	90.91	22.44	50.59			
Specificity (%)	99.73	99.78	99.43			
True switch	26	9	2	0	5	1

Table 3

Results of the algorithm for chest electrode reversals; see text for explanation of the legends.

	V1–V2	V2–V3	V3–V4	V4–V5	V5–V6
True negative	18,405	18,391	18,360	18,396	18,400
False positive	19	33	64	28	24
True positive	13,792	15,236	17,995	17,917	16,817
False negative	4632	3188	429	507	1607
Sensitivity (%)	74.86	82.70	97.67	97.25	91.28
Specificity (%)	99.90	99.82	99.65	99.85	99.87
True switch	10	48	16	37	81

was detected only in 22% of the cases, but with a reasonable specificity. The diagonal switch right arm–left leg has a 51% sensitivity and a 99.4% specificity, which was the lowest of all combinations analyzed. The reversals that involve the right leg (neutral) electrode have very little effect, but do not generate false alerts either.

The false positive cases would give rise to a “nuisance alert” at the electrocardiograph. Too many nuisance alerts will cause the technician to ignore the lead reversal message eventually, rendering the algorithm useless. It is therefore important to keep the specificity high, in particular for the low occurrence cases. The right arm–left leg reversal is the biggest culprit for false alerts (31% of all false alerts), and does little good: the number of real RA–LL electrode reversals is low. Apparently, this diagonal switch is an improbable mistake: given the detection rate of 51%, we can estimate it at one in every 5000 ECGs taken at this institution. Also, switches involving the right leg electrode are rare. In principle, these reversals are not too difficult to detect by an automatic program, because their symptom is a flat lead I, II or III; however, very few true positive cases were seen. It is likely that ECG technicians notice a completely flat lead and check for this kind of error. Curiously, Hedén [5] found many more RA–RL reversals in a database from the emergency room, although their total reversal rate is similar to that in our set.

In order to reduce the number of false positives for rare reversals, it was decided to remove the diagonal limb electrode reversals from the algorithm. The final result of the remaining reversals can be seen in Table 4, together with the results of Hedén et al. [4,5] and Kors [6].

Table 4

Overall performance and comparison between our study and those of Hedén et al. [4,5] and Kors and Van Herpen [6].

	RA–LA	LA–LL	V1–V2	V2–V3	V3–V4	V4–V5	V5–V6	ALL
Sensitivity								
De Bie	90.91	22.44	74.86	82.70	97.67	97.25	91.28	79.59
Hedén	97	57.6	80.6	44.5	77.5	83.0	73.2	73.3
Kors	98.6	17.9	94.3	87.4	97.5	99.1	94.0	84.11
Specificity								
De Bie	99.73	99.78	99.90	99.82	99.65	99.85	99.87	98.60
Hedén	99.94	99.97	99.94	99.87	99.95	99.95	99.88	99.5
Kors	100	99.5	100	99.9	99.7	99.7	100	98.8

For the RA–LA sensitivity from the Hedén study, the weighted average of the performance with and without P-wave criteria is reported. The “ALL” sensitivity is the average of the separate sensitivities (which presumes the same prior probability of all reversals), while the “ALL” specificity is calculated as the sum of the 7 specificities minus 600%.

Although the algorithm was tested retrospectively, it was designed and implemented in electrocardiographs in real time. An alert is given to the technician for a possible electrode reversal before the ECG is taken and processed. Based on the result of this study, the total false positive alert rate is 1.4%, or one false alert in every 70 ECGs, which should not be a significant bother to an ECG technician.

The fraction of alerts that is actually true depends on the accuracy of the technician. At the site where the database in this study was taken from, the number of electrode reversals can be estimated at 1.5%, considering the detection sensitivity of the algorithm. At this site, the electrocardiograph would give an electrode reversal warning once every 35 ECGs taken, and about half of them would be true. It can also be estimated that the reversal rate at this site would be reduced to 0.3% with the discussed algorithm, a considerable improvement.

When comparing the results of the various studies, it is important to consider the providence, composition and number of the ECGs used. We chose not to include Xia's [7] data in the comparison, because the datasets used were not representative for regular hospital ECGs, and the number of records from different patients was low (total 705 patients from 3 different datasets), insufficient for statistical reliability. Also the dataset Kors and Van Herpen [6] used was small, 3305, of which 1220 were from a set acquired for a specific study, ECGs with high technical quality, without bundle branch blocks. The other ECGs came from the cardiology ward of a university hospital. Two thirds of these ECGs were used as a learning set, only 1102 as a test set to compute the results. All ECGs were manually screened for true electrode reversals, which turned out to be 1.1%.

Hedén's dataset is considerably bigger; it consists of about 11,000 ECGs from the emergency department of a university hospital. For each reversal, the set was divided into 2 parts, a training set and a test set. The true reversal rate in this database was 1.8%. Both Hedén and Kors, like us, excluded pacemaker ECGs from the study. The number of ECGs excluded for this reason was 1.8% and 1.4% respectively.

The database used in our study was the biggest of these, over 18,000. They came from all departments of the hospital, over the course of almost 10 years. Over time, two generations and several models of ECG devices were used. The procedure for selecting only ECGs for which there were three or more ECGs in the database, probably has caused some bias for "sicker" patients, and more abnormal ECGs. An indication for this is the relatively high number of pacemaker ECGs that was excluded (4.5%), more than double than those found in the previous studies. Given the low prevalence of individual electrode reversals, and the high specificity required for any algorithm to be useful, such a high number of ECGs is necessary for statistical reliability. The use of two serial ECGs was essential for the establishment of the truth of the positive reversal results. Clearly, with such a high number of ECGs, it was not possible to establish the truth of the negative reversal outcomes. However, using the simulated reversals, we know

the sensitivity of the algorithm for each; knowing also the fraction of true positives, we can estimate the number of false negatives, and thus the prevalence of electrode reversals in the original dataset. The ECGs used for the development of the criteria for our algorithm were not part of the database used for testing.

The performance of our algorithm is similar to Kors and Van Herpen [6], while the overall specificity of the Hedén et al. [4,5] method is somewhat higher and sensitivity somewhat lower. Although all ECG interpretations contain various electrode reversal detection algorithms, as far as we know, our algorithm is the only one implemented in electrocardiographs in real time, that is, giving a warning to the technician before the ECG is captured. Although performed with little data and very dissimilar databases, Xia's paper [7] suggests that both the Kors and the Hedén results are difficult to reproduce and are worse on datasets dissimilar from the ones that they were learned with.

Conclusion

An electrode lead reversal algorithm was developed and tested for use in the practice of routine resting ECG acquisition for the purpose of alerting the technician to the problem before the ECG is taken. The algorithm detects the most common limb electrode reversals as well as all neighboring chest electrode reversals. The algorithm was tested on a large dataset representative of ECGs taken in all departments of a hospital. Based on the results, it is expected that the number of lead reversals in routine practice can be reduced by circa 80% with the algorithm, with a modest false alert rate of circa 1.4% of the ECGs taken.

Acknowledgments

The ECGs used in this study have been collected and made available in anonymous form by the Azienda Ospedaliera Universitaria Santa Maria della Misericordia in Udine, Italy by mutual agreement between the University and Mortara Instrument, Inc.

References

- [1] Batchvarov VN, Malik M, Camm AJ. Incorrect electrode cable connection during electrocardiographic recording. *Europace* 2007;9:1081–90.
- [2] Rudiger A, Schöb L, Follath F. Influence of electrode misplacement on the electrocardiographic signs of inferior myocardial ischemia. *Am J Emerg Med* 2003;21:574–7.
- [3] Badri M, Goldstein MA, Burke JF, Kowey PR. Reversal of fortunes: recognizing misplacement of ECG electrodes. *Am J Med* 2013;10:868–70.
- [4] Hedén B, Ohlsson B, Edenbrandt L, Rittner R, Pahlm O, Peterson C. Artificial neural networks for recognition of electrocardiographic lead reversal. *Am J Cardiol* 1995;75:929–33.
- [5] Hedén B, Ohlsson B, Holst H, Mjöman M, Edenbrandt L, Rittner R, et al. Detection of frequently overlooked electrocardiographic lead reversals using artificial neural networks. *Am J Cardiol* 1996;78:600–4.
- [6] Kors JA, Van Herpen G. Accurate automatic detection of electrode interchange in the electrocardiogram. *Am J Cardiol* 2001;88:396–9.
- [7] Xia H, Garcia GA, Zhao X. Automatic detection of ECG electrode misplacement: a tale of two algorithms. *Physiol Meas* 2012;33:1549–61.